AI Development Log — CollabBoard

Tools & Workflow

ChatGPT was used for presearch and architectural reasoning. Claude acted as the primary reasoning engine for feature design and higher-level logic. Cursor was used as the implementation layer — effectively the execution environment where AI-generated code was written, debugged, and refined. Workflow: presearch → feature planning in Claude → implementation and debugging in Cursor. No hand-written feature code was produced by the developer.

MCP Usage

No MCP servers were used. All development occurred through direct AI interactions inside Claude and Cursor without external tool integrations.

Effective Prompts

Presearch Prompt:

```
Given these requirements and Appendix for presearch let's get started. Let's review the requirements and discuss this topic. Ask me questions along the way while doing this Presearch.
```

Auth State Machine Prompt:

```
Let's check the state machine. If sign up is valid it should show something like sign up complete and take them to sign in. If correct credentials are placed it should take them to the app where they can start using it.
```

Style Bar Prompt:

```
I want to copy Miro's style bar concept. I am in Cursor — give me a prompt so Cursor can successfully create the top style bar. When something is selected in the style bar I want: option to switch out shapes, the color, and if it is a sticky note or text box I want to see font B/U/I, text size. I want to see x and y coordinates.
```

Layering Prompt:

```
The layering of shapes are off — the one that is generated last stays on top. Either have the AI take that into consideration or fix the layering so that based on how I drag a shape it puts it on top, last clicked goes on top or something like that.
```

Bulk Generation Prompt:

```
Make a tool call for the AI so that it can add multiple shapes at once ex 'add 500 rectangles' it generates 500 rectangles apply it for all shapes
```

Code Analysis

Metric	Value
Total Project Code	~18,700 lines
Test Code	~4,400 lines (39 files)
Total Commits	60
Estimated AI-Generated Code	≈100% (developer only modified .env configuration)

Strengths & Limitations

Strengths: Rapid scaffolding, full feature implementation, strong architectural reasoning, and ability to iterate quickly. AI handled infrastructure, UI features, collaboration logic, tool systems, and refactors efficiently.

Limitations: Testing required manual validation. UI/UX quality still needed human judgment. Context switching between AI agents occasionally introduced merge conflicts.

Key Learnings

Prompt precision directly impacts output quality. Providing hypotheses (not just error logs) improves debugging results. Separating reasoning (Claude) from execution (Cursor) reduces friction. Polishing and edge-case handling take longer than initial feature generation. AI is extremely effective at building systems, but structured oversight is essential.